A Downlink Pilot Based Signal Processing Method for Integrated Sensing and Communication Towards 6G

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***Abstract*—Integrated Sensing and Communication (ISAC) is an emerging technology that realizes both communication and sensing functionalities on the same sets of hardware as well as same frequency bands. Such technique will bring tremendous benefits such as spectrum efficiency improvement and cost reduction. In this paper, we firstly define three different classes of ISAC with the integration of both resources and functionalities for communication and sensing as the ultimate form. Furthermore, target at the full-integration level, we propose a downlink-pilot-based OFDM ISAC signal processing algorithm for base-station-orientated sensing. The OFDM architecture guarantees forward compatibility, and downlink reference signals including DMRS and CSI-RS ensure the communication throughput and address privacy concerns as well. Simulation results show that the proposed method is able to achieve outstanding sensing performance with range estimation error of 0.15m and velocity estimation error of 0.2m/s in the low SNR region.**

Keywords—6G, Integrated Sensing and Communication, Joint Sensing and Communication, Integrated Signals, OFDM

I. INTRODUCTION

As cellular networks are evolving to operate on higher frequency bands, the communication spectrum and radar spectrum will converge and lead to interference issues. Since the signal processing structure of communication and radar sensing systems are becoming more similar as well, the integration of communication and sensing is an evitable trend for future 6G network. Integrated Sensing and Communication is a novel information processing framework that achieves cooperation and integration of different functions based on hardware and software resource sharing. Through the signal co-design and hardware sharing, the integrated sensing and communication technique will be able to acquire sensing information (such as orientation, distance, velocity, etc.) during the process of information transmission. Such an approach will be able to conduct detection, tracking, imaging and other functions of target devices, events and/or environments, which leads to the overall performance improvement. In this introduction section, we will first define three levels of integration, followed by the literature review as well as main contribution and structure of this paper.

A. Different levels of ISAC

There are three categories of integration, defined from the perspective of integration of resources and integration of functions, namely:

1) Sensing and communication functions are integrated while their resources are independent.

In this category, sensing and communication operate on separate hardware and spectrum resources. The integration of functions is based on information sharing, meaning that sensing (communication) information can be used to assist communication (sensing).

2) Sensing and communication resources are integrated while their functions are independent.

In this group, the functionalities of sensing and communication are separate whereas their spectrum and hardware resources are integrated. The integration of hardware includes sharing of the same set of antennas, RF components, baseband units, etc.

3) The functions and resources of sensing and communication are both integrated.

This class represents the ultimate goal of integration, where same spectrum and hardware resources are used to simultaneously realize both sensing and communication functions. Through the same hardware and software platforms, sensing and communication will be deeply integrated and become the key feature of 6G networks. Such a system will enable mutual enhancement of sensing and communication by actively sensing and analyzing the wireless channel characteristics while transmitting information, so as to picture the physical characteristics of the surrounding environment.

B. Literature Review

There has been a lot of work on ISAC, and we will give a brief overview in the following.

In terms of theoretical boundaries of ISAC systems, [1] defined the radar estimation information rate using the Cramér-Rao lower bound of the time-delay estimation, and derived the different performance inner bounds such as successive interference cancellation inner bound, water-filling inner bound and optimal fisher information inner bound. Kobayashi *et al.* derived the achievable capacity-distortion region for memoryless state-dependent multiple access channel which relates to the ISAC system [2]. Chiriyath *et al.* introduced a novel information matrix that combines the communication data rate and the radar estimation rate [3]. Besides, in [4], the authors defined the radar capacity as the number of distinguishable targets which is also the maximum information carried by the radar echo signal.

As a major use case of ISAC, the vehicle-involved scenario is a popular research topic. Based on the IEEE 802.11ad standards, [5] utilized the good correlation properties of the Golay complementary sequences for object detection, distance and velocity estimations. Similarly, [6] used IEEE

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802.11p standards for integrated design, and proposed a forward detection and backward communication method for the vehicular ad hoc networks, and gave the expression of upper bound for the successful transmissions. In [7], the impact of spectrum sharing principles among communication and sensing are discussed under the multi-vehicle cooperative detection setting. Furthermore, D. Ma *et al.* provided thorough discussion for the waveform selection and wireless resource distribution, as well as the benefits and drawbacks of implementing ISAC in the autonomous driving systems [8].

The integrated signal design is an indispensable topic for the ISAC system, which allows the full integration where the same time and frequency resources are used for both sensing and communication purposes. There are two major directions on the integrated signal design. One is based on the radar waveform (such as linear frequency modulation signal, also known as LFM or chirp signal), and the other is based on the existing communication waveform such as OFDM. For the radar-waveform-based methods, the author in [9] modulated the communication information onto the LFM signals where bit 0 and bit 1 are modulated on the up-chirp and down-chirp signals respectively. Gaglione *et al.* used fractional Fourier transform for LFM signals to be modulated by the communication data, where a low bit-error-rate is achieved while assuring the sensing performance [10].

On the other hand, regarding the communication-oriented waveform design, [11] evaluated the performance of both communication and radar imaging when using OFDM signals on the airborne platforms. In addition, in [12], a software-defined OFDM radar sensing and communication system is proposed for finishing ranging, imaging and communication tasks. Another important set of work is [13]-[15], where the periodogram-based methods are used to process the reflected the OFDM communication data signals to get sensing parameters. Accordingly, in [15], the authors finished the field test in the highway vehicle-to-vehicle scenario with the verification of vehicle detection and motion parameter estimation.

C. Main Contribution and Structure of the Paper

Current literatures have their own limitations and cannot meet the requirements of base-station-based (BS-based) sensing. For instance, [5][6] used IEEE 802.11standards and [9][10] utilized chirp signals which are not compatible with cellular networks. In [13]-[15], the proposed method was to use OFDM data signal for sensing in the vehicle-to-vehicle detection and ranging scenario, where no base stations were involved and the transceiver was equipped on the vehicles. Moreover, privacy and data security issues arise if we use this method to exploit data signals in the BS-based sensing scenarios.

Therefore, in this paper, we propose a downlink reference signal based OFDM ISAC signal processing method using two-dimensional Fourier operations. The benefits of using downlink reference signals is that their information are pre-defined and known by different access points and devices. In addition, data security and privacy concerns are also addressed. The main contribution of this paper are:

1) We propose the downlink pilot based OFDM ISAC signal processing methods for BS-based sensing built upon on [15]. We utilize demodulation reference signals (DMRS) and channel state information reference signals (CSI-RS) for sensing, which are universally used as downlink pilots. Through using downlink pilots for sensing, the positioning reference signal (PRS) could be abandoned in 6G systems.

2) We investigate how different time-frequency resource patterns of the reference signals influence distance and velocity estimation accuracy.

3) We give the resolution and accuracy derivations for range and velocity estimation. We also investigate the method of improving accuracy by increasing number of Fourier operation points, and conduct simulations accordingly.

The rest of the paper is organized as follows: Section II will define the system model. Section III will introduce the proposed reference-signal-based signal processing, together with the discussion on resolution and accuracy. Section IV will demonstrate the simulation results, and analyze the performance accordingly. Section V will conclude this work and provide some potential future research directions.

II. SYSTEM MODEL

We consider the BS-based scenario where the transmitter and the receiver are equipped on the same base station. We assume that the target objects are located within the BS coverage, and the BS is able to receive the echo signal reflected from the target. The target object does not necessarily have established a connection with the BS. At the transmitter side, the time-domain OFDM signal is expressed as

$$x(t) = \sum_{l=0}^{L-1}\sum_{k=0}^{K-1} S(lK+k)\exp(j2\pi f_k t) \quad 0 \le t \le T \quad (1)$$

where K is the total number of subcarriers, and L *is* the total number of OFDM symbols, f_k is the subcarrier frequency. $S(k,l) = S(lK+k)$ is the modulation symbol corresponds to the downlink reference signals at the lth subcarrier and k^{th} OFDM symbol.

Assuming the distance between the targeted object and transmitter is R, the velocity of the target is v relative to the transmitter, the reflected signal captured at the receiver end is

$$y(t) = \sum_{l=0}^{L-1}\sum_{k=0}^{K-1} S(k,l)exp(j2\pi f_k t) \times exp\left(-j2\pi k\Delta f\frac{2R}{c}\right)exp\left(j2\pi\frac{2v}{\lambda}t\right) + n(t) \quad (2)$$

where $n(t)$ is the additive Gaussian white noise with zero-mean and unit-variance. Equation (2) can be re-organized as

$$y(t) = \sum_{l=0}^{L-1}\sum_{k=0}^{K-1} S_r(k,l)exp\,(j2\pi f_k t) + n(t) \quad (3)$$

with the substitution of

$$S_r(k,l) = S(k,l)\,exp\left(-j2\pi k\Delta f\tfrac{2R}{c}\right)exp\left(j2\pi\tfrac{2v}{\lambda}t\right) \quad (4)$$

where c is the speed of light and λ is the wavelength of the signal. From (4) we can derive that the received modulation symbol $S_r(k,l)$ is the channel-influenced version of the transmitted modulation symbol $S(k,l)$, where the channel introduced a delay of $\tau = 2R/c$ and Doppler shift of $f_D = 2v/\lambda$ [15]. Therefore, if we perform the element-wise

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division of received and transmitted modulation symbols of downlink pilots, the transfer function of the channel $D(k,l)$ which contains the delay and Doppler information can be found as follows

$$D(k,l) = \frac{S_r(k,l)}{S(k,l)} \tag{5}$$

By conducting the signal processing of the channel transfer function, we can obtain the delay and Doppler information of the channel and further acquire the distance and velocity parameters of the target.

III. Downlink-Pilot-Based OFDM ISAC Signal Processing

In this section, we will introduce the reference-signal-based OFDM ISAC signal processing structure, with discussions of resolution and accuracy at the end.

A. OFDM ISAC Signal Processing for Downlink Pilots

The essence of the OFDM ISAC reference signal processing algorithm is first take the inverse Fourier transform of channel transfer function (5) along the frequency axis and Fourier transform of it along the time axis [14]. Then find the two-dimensional index corresponds to the maximum values of the Fourier operation results, and calculate the target distance and velocity accordingly. Fig. 1 demonstrates downlink-pilot-based OFDM ISAC signal processing process. In the following, we will discuss how the channel transfer function (5) can be used for distance and velocity estimation in more details.

Given the target distance R, the subcarriers corresponds to the same individual OFDM symbol will experience the linear phase shift caused by the delay $\tau = 2R/c$. Assume the target is stationary, and the channel transfer function in this scenario is

$$D(k) = exp\left(-j2\pi k\Delta f\frac{2R}{c}\right) \tag{6}$$

By taking the inverse Fourier transform of (6), we have

$$h(m) = IFFT\{D(k)\} = \frac{1}{K}\sum_{k=0}^{K-1} D(k)\exp\left(j2\pi\frac{k}{K}m\right)$$
$$= \frac{1}{K}\sum_{k=0}^{K-1} exp\left(-j2\pi k\Delta f\frac{2R}{c}\right)\exp\left(j2\pi\frac{k}{K}m\right) \tag{7}$$

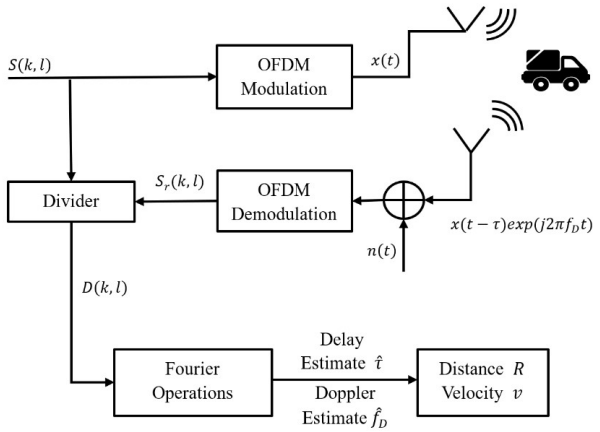


Fig. 1. Signal processing algorithm for downlink reference-signal-based OFDM ISAC systems.

When the exponential terms are canceled out, $h(m)$ reaches its maximum. By finding the index m_{max} corresponds to the maximum value of $h(m)$, the target distance R can be obtained as follows

$$R = \frac{m_{max}c}{2K\Delta f} \tag{8}$$

Similarly, the moving object will introduce the linear phase shift corresponds to Doppler frequency $f_D = 2v/\lambda$ onto a given subcarrier. If we only consider the effect brought by the target velocity, the channel transfer function can be written as

$$D(l) = exp\left(j2\pi\frac{2v}{\lambda}lT_{OFDM}\right) \tag{9}$$

where T_{OFDM} is the duration of an OFDM symbol. Through taking Fourier transform of (9), we can get

$$h(n) = FFT\{D(l)\} = \sum_{l=0}^{L-1} D(l)\exp\left(-j2\pi\frac{l}{L}n\right)$$
$$= \sum_{l=0}^{L-1} exp\left(j2\pi\frac{2v}{\lambda}lT_{OFDM}\right)\exp\left(-j2\pi\frac{l}{L}n\right) \tag{10}$$

With a similar spirit of analyzing (7), when the exponential terms of (10) cancel each other, $h(n)$ will reach its maximum, and the target velocity can be acquired by finding the index n_{max} relates to the maximum value of $h(n)$

$$v = \frac{n_{max}\lambda}{2LT_{OFDM}} \tag{11}$$

B. Resolution and Estimation Accuracy

In this subsection, we will go through the resolution and accuracy analysis. From (8), we can easily derive the range resolution

$$\Delta R = \frac{c}{2K\Delta f} \tag{12}$$

In the radar sensing system, the resolution is inversely proportional to the bandwidth, namely $\Delta R = c/2B$. If we assume that the system bandwidth is the product of the subcarrier spacing and the number of subcarriers, i.e., $B = K\Delta f$, we can reach the same result of (12).

From (11), we can get the velocity resolution

$$\Delta v = \frac{\lambda}{2LT_{OFDM}} \tag{13}$$

In the radar sensing system, the velocity resolution is defined as $\Delta v = \lambda/2\tau$ where τ is the signal duration time which can be also expressed as the product of the OFDM symbol duration and number of OFDM symbols LT_{OFDM}. Substitute $\tau = LT_{OFDM}$ into $\Delta v = \lambda/2\tau$, we can also get (13). The above derivations show that the resolution analysis of OFDM ISAC algorithms is in line with that of the radar systems.

One thing that was not explicit but should be noted is that we assume that the number of FFT/IFFT points in (7) (10) is the same as the number of subcarriers or the number of OFDM symbols. However, in the existing OFDM wireless cellular networks, the number of FFT/IFFT points is the positive integer power of two (i.e. 2^P), which would be larger than the number of subcarriers or the number of OFDM symbols. Therefore, regarding the range-related expressions, the number of subcarriers K should be replaced by the number of IFFT points in (8)(12). In terms of velocity-related expressions, the number of OFDM symbols L should be replaced by the number of FFT points M in (11)(13).

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Therefore, the equations related to distance and velocity should be re-written as

$$R = \frac{m_{max} c}{2N\Delta f} \tag{14}$$

$$\Delta r' = \frac{c}{2N\Delta f} \tag{15}$$

$$v = \frac{n_{max}\lambda}{2MT_{OFDM}} \tag{16}$$

$$\Delta v' = \frac{\lambda}{2MT_{OFDM}} \tag{17}$$

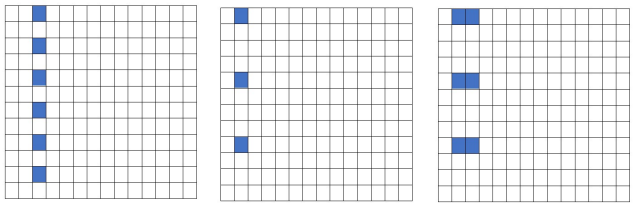
It should be highlighted that $\Delta r'$ here is not range resolution which is a sensing-system indicator of the ability to distinguish closely-located targets. As the range resolution is inversely proportional to bandwidth, the range resolution is a constant quantity for a given bandwidth. Hence, the $\Delta r'$ in (15) is no longer the resolution, but actually represents the quantization interval brought by Fourier operations. This indicates that $\Delta r'$ characterize the range accuracy (or estimation error) brought by quantization instead of the ability to differentiate the nearby objects. Similarly, $\Delta v'$ is not the velocity resolution either. Observing (15) (17), we can find that the range and velocity estimation accuracy can be improved by increasing the number of FFT/IFFT points.

IV. Simulation Results and Discussions

This section will present the simulation results and analysis accordingly. Table I illustrates the simulation parameters where mmWave band is used. We conducted Monte Carlo simulation 100 times for each simulation setting. In order to compare different patterns associated with different reference signals, three patterns are used as shown in Fig.2. The first pattern is a specific DMRS pattern, the latter two are two specific CSI-RS patterns. Details regarding three different patterns will be discussed later.

TABLE I. Simulation Parameters

Parameters	Value
Carrier Frequency	26GHz
Subcarrier Spacing	120kHz
Number of Slots	10
Number of Resource Blocks	264
Modulation Scheme	QPSK
Target Distance	300m unless otherwise stated
Target Velocity	20m/s unless otherwise stated



(a) (b) (c)

Fig. 2. Different resource patterns with the horizontal axis representing time and the vertical axis representing frequency: (a) a specific DMRS pattern, (b) a specific CSI-RS pattern, (c) another specific CSI-RS pattern.

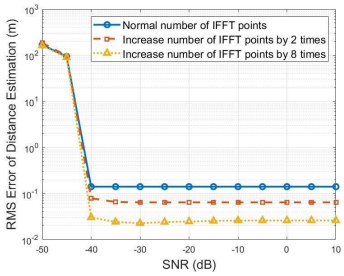


Fig. 3. RMS error of distance estimation for different number of IFFT points.

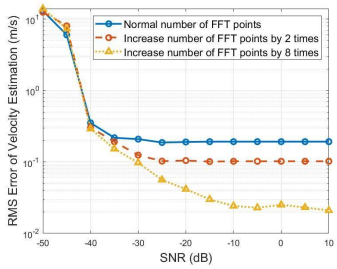


Fig. 4. RMS error of velocity estimation for different number of FFT points.

Fig. 3 and Fig.4 demonstrate the RMS error of the 100 times Monte Carlo simulations for the distance and velocity estimation respectively, when using the DMRS pattern in Fig.2(a) and applying different number of FFT/IFFT points. From the results we can see that in the convergence state, when the number of FFT/IFFT points used increases, the RMS error decreases. As discussed in Section III, when the number of points for Fourier operations increases, the quantization interval becomes smaller and leads to better accuracy. This is also evident in Fig.3 and 4, where the RMS error is decreased approximately 2 and 8 times when the number of FFT/IFFT points are increased by 2 and 8 times respectively. However, the use of more FFT/IFFT points will introduce computational complexity and is a trade-off to consider in different scenarios. Fig.3 and 4 also demonstrate excellent sensing capabilities with the distance estimation error as 0.15m and the velocity estimation error as 0.2m/s even in the low SNR region (from above -30dB).

Fig. 5 and Fig. 6 show RMS error for the distance and velocity estimation respectively, with the comparison of three different patterns in Fig. 2. As the proposed method operates on the modulated symbol, thus the difference in the transmitted data stream will not influence the algorithm performance. This means that DMRS and CSI-RS will have the same sensing performance if the time-frequency resource patterns are the same regardless of the different sequences that are used. Simulation results also prove this, the corresponding result figure is not shown here for the conciseness of the paper. In Fig.2, pattern(a) occupies 6 and pattern(b)(c) occupy 3 subcarriers in each research block (RB). Regarding the time-domain, pattern(a)(b) occupy one OFDM symbol in each slot and pattern(c) uses two.

In Fig.5, pattern(a) have higher errors compared with the other two patterns which have the same error value at convergence state. This is because pattern(b) and (c) have the same frequency granularity, and are sparse than pattern(a). Therefore, in (15), Δf becomes the effective subcarrier

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spacing, and pattern(a) has smaller effective subcarrier spacing compared with pattern(b)(c), which leads to higher RMS error eventually. In Fig.6, three patterns have the same velocity estimation error in the convergence state. This is because the OFDM symbol duration T_{OFDM} in (17) for three patterns are the same although the time domain densities are different.

Furthermore, we conduct multi-target range and velocity estimation, Fig.7 demonstrates the 2D radar sensing image of 5 targets. The distance and velocity estimates are correctly shown in the figure, which demonstrates the multi-target sensing capability of the proposed method.

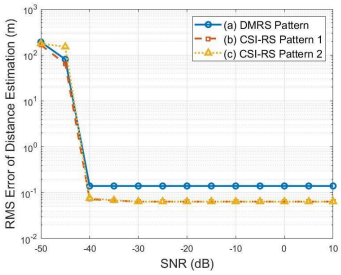


Fig. 5. RMS error of distance estimation for different resource patterns.

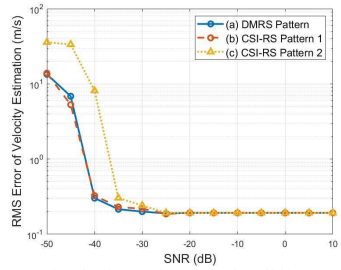


Fig. 6. RMS error of velocity estimation for different resource patterns.

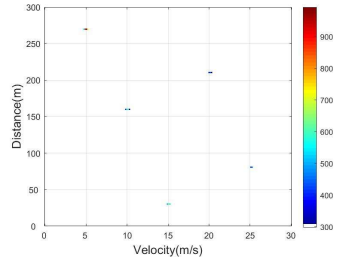


Fig. 7. Radar image for 5 targets with distance and velocity of [270m, 5m/s], [30m, 15m/s], [160m, 10m/s], [80m, 25m/s], [210m, 20m/s] respectively.

V. CONCLUSION

In this paper, we first define three different class of ISAC with the integration of both resources and functionalities as the fully-integrated level. Aim at the full integration of communication and sensing, we propose a downlink pilot based OFDM ISAC signal processing method using two-dimensional Fourier operations. We utilized downlink reference signals (including DMRS and CSI-RS) to ensure the communication throughput and address privacy concerns. Simulations results show that the proposed method is able to achieve outstanding sensing performance with range estimation error of 0.15m and velocity estimation error of 0.2m/s. We also investigated the means to improve sensing accuracy, including increasing the number of Fourier operation points but at a cost of computational complexity. We also discussed how different time-frequency resource patterns affect the sensing error. Regarding future research directions, since in this work we consider the transmitter and receiver are equipped on the same base station, the interference among sensing signals as well as interference between sensing and communication signals are not highlighted and should be further investigated. The implementation of multi-BS cooperation and networking could play an important role in interference management.

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